



DPUSGEDistortionQuad

Distorts the image of a SGEView to an arbitrary quadrilateral.

Distorts the image of a SGEView using two-dimensional Bezier interpolation.

1. Dynamic parameters:

<i>XTL, YTL</i>	Coordinates (in the 2D overlay coordinate system) where the top-left edge of the scenery is placed.
<i>XTR, YTR</i>	Coordinates of the top-right edge of the scenery.
<i>XBL, YBL</i>	Coordinates of the bottom-left edge of the scenery.
<i>XBR, YBR</i>	Coordinates of the bottom-right edge of the scenery.
<i>NDX, NDY</i>	Coordinates (in the 2D overlay coordinate system) where the undistorted top-left edge of the scenery is additionally placed, if the additional undistorted output is activated (see below)
<i>NDW, NDH</i>	Width and height of the additional undistorted output
<i>TexSMin, TexSMax</i>	Specifies which range of the computed image (in horizontal dimension) is actually displayed. Default: (0...1), i.e. the whole computed image is displayed.
<i>TexTMin, TexTMax</i>	Specifies which range of the computed image (in vertical dimension) is actually displayed. Default: (0...1), i.e. the whole computed image is displayed.

2. Static parameters:

<i>ViewID</i>	The number of the view whose image is distorted. Counting starts at 0, which is assigned to the first view that is listed in the configuration.
<i>ResolutionX, ResolutionY</i>	Size of the off-screen image buffer (in pixels), into which the scenery is rendered. Should correspond to the image size after distortion. Large Resolution values reduce the achievable frame rate. Note: For older graphics boards (e.g. NVidia GeForce 6000, 7000 series), powers of two must be used for ResolutionX and ResolutionY, otherwise no output can be seen. Newer boards can use arbitrary values.
<i>SubdivX, SubdivY</i>	Subdivision of the projection geometry in X and Y direction. The more the image is distorted, the more subdivisions are required; however, that reduces the achievable frame rate. The default value is 50 and is sufficient for most practical purposes.
<i>DistortResetViewport</i>	Selects whether the coordinates XTL ... YBR and NDX ... NDH are relative to the whole window into which the view is rendered (1) or only to the selected viewport (0).



The default value is 0 and only needs to be changed if an additional undistorted output is desired. (See below)

3. Dependencies to other DPUs:

DPUSGEDistortionQuad must run together with **DPUSGEWorld** and **DPUSGEView**.

The following Index values must be assigned:

DPUSGEWorld.Index < DPUSGEDistortionQuad.Index < DPUSGEView.Index

4. Functional principle:

The DPU works well with the following script:

```
DPUSGEDistortionQuad SGEDistortion
{
    Computer = {VIS};
    Index = 50;

    ResolutionX = 1024;
    ResolutionY = 512;

    XTL = 150;
    YTL = 0;

    XTR = 1024;
    YTR = 150;

    XBR = 900;
    YBR = 768;

    XBL = 0;
    YBL = 630;
};
```

Assuming that the associated SGEView has a resolution of 1024 x 768 pixels, the scenery is rotated slightly to the left using these parameters. The result is displayed in the following screenshot.



Figure 1: Scenery distorted by DPUSGEDistortionQuad

5. Additional undistorted output:

In many cases, it is useful to display not only the distorted scenery (on a projector), but additionally the undistorted scenery (on an observation monitor that has been connected in parallel).

DPUSGEDistortionQuad supports this configuration without a noticeable performance loss.

The following setup is required for this configuration:

- The projector and the computer monitor are connected to the same computer and to the same graphics board.
- Both displays use the same resolution (e.g. 1400x1050 pixels each).
- In the NVidia Control Panel (or the AMD Catalyst Control Center), the multi-monitor option is set to "Horizontal span".

The next step is to configure DPUSGEView such that it creates an output window which spans both displays. However, the scenery is – at first – rendered on only one of the displays:

DPUSGEView SGEView



```
{  
    Computer = {VISF01};  
    Index = 80;  
    ...  
    WindowW = 2800;  
    ViewportX = 0;  
    ViewportY = 0;  
    ViewportW = 1400;  
    ViewportH = 1050;  
};
```

Finally, **DPUSGEDistortionQuad** is added to the configuration. It will output a distorted image on the projector and an undistorted image on the observation monitor.

DPUSGEDistortionQuad **SGEDistortion**

```
{  
    Computer = {VISF01};  
    Index = 50;  
  
    ResolutionX = 1400;  
    ResolutionY = 1050;  
  
    XTL = 150;  
    YTL = 0;  
  
    XTR = 1400;  
    YTR = 150;  
  
    XBR = 1250;  
    YBR = 1050;  
  
    XBL = 0;  
    YBL = 900;  
  
    DistortResetViewport = true;  
  
    NDX = 1400;  
    NDY = 0;  
  
    NDW = 1400;  
    NDH = 1050;  
};
```

The result is displayed in the following figure.



SILAB DRIVING SIMULATION

version Version 7.1 documentation

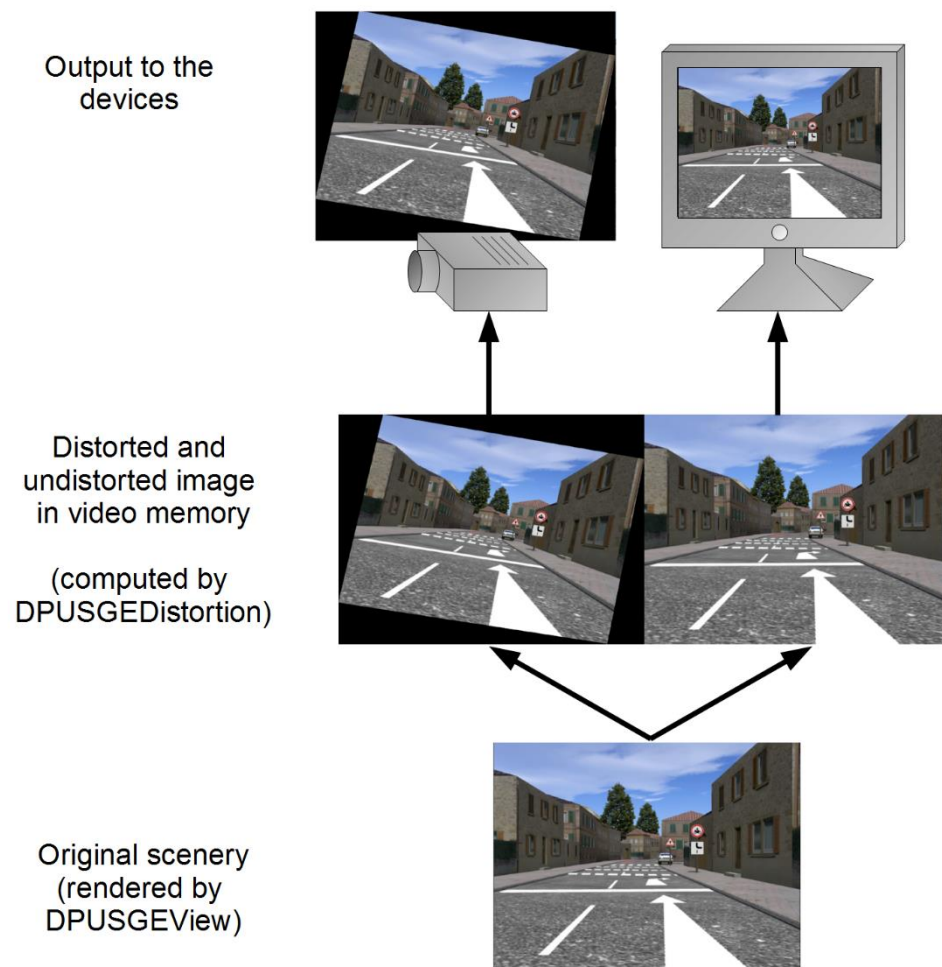


Figure 2: Parallel output of distorted and undistorted image